



SOP Title:
Zooarchaeology by Mass Spectrometry (ZooMS)
FOR BONE MATERIAL

Document Number: #ZooMS02

SOP Originator: ZooMS PhD Student	Approving Position: Palaeoproteomics Lab Head	Effective Date: 04/Feb/2019
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LOCATION:

Drilling Room, ZooMS Wet Chemistry Lab, ZooMS MALDI Lab

PRINCIPAL:

Digestion of fragments of archaeological or modern bone proteins for species identification via MALDI TOF mass spectrometry.

COSHH RE

Risk assessment is evaluated by the Lab Manager/Safety Officer. The safety data sheets pertaining to the lab are in an extra folder stored in the lab and all details about hazardous goods or other chemical components can be found in the database: <https://www.dguv.de/ifa/gestis/gestis-stoffdatenbank/index.jsp>.

TRAINING REQUIRED

Centrifuging, Pipetting, Balance, Sonicator

ZooMS Wet chemistry training, MALDI TOF training

Please see *Krisitne Richter* and *Jana Zech* or *Sandra Hebestreit* (Technicians) prior to starting any work detailed in this SOP if you have not completed all of the training listed above.



IF YOU USE THIS FOR PUBLICATION PLEASE CITE:

Buckley, Michael, Matthew Collins, Jane Thomas-Oates, and Julie C. Wilson. 2009. "Species Identification by Analysis of Bone Collagen Using Matrix-Assisted Laser Desorption/Ionisation Time-of-Flight Mass Spectrometry." *Rapid Communications in Mass Spectrometry* 23: 3843–54.

If using Non Destructive method (Section A) also cite:

van Doorn, Nienke Laura, Hege Hollund, and Matthew J. Collins. 2011. "A Novel and Non-Destructive Approach for ZooMS Analysis: Ammonium Bicarbonate Buffer Extraction." *Archaeological and Anthropological Sciences* 3(3): 281–89.

If using the Acid Insoluble method (Section B) also cite:

Welker, Frido et al. 2015. "Using ZooMS to Identify Fragmentary Bone from the Late Middle/Early Upper Palaeolithic Sequence of Les Cottés, France." *Journal of Archaeological Science* 54: 279–86.

If using the Acid Soluble method (Section C) also cite:

van der Sluis, L G et al. 2014. "Combining Histology , Stable Isotope Analysis and ZooMS Collagen Fingerprinting to Investigate the Taphonomic History and Dietary Behaviour of Extinct Giant Tortoises from the Mare Aux Songes Deposit on Mauritius." *Palaeogeography, Palaeoclimatology, Palaeoecology* 416: 80–91.



I SECTION: REAGENT RECIPES/INSTRUCTIONS SHEET

The usage of a fume hood for preparing HCl, TFA, and ACN containing solutions is mandatory. All final reagent bottles or tubes should have the name of the solution, concentration of solution, name of the person who prepared it, and date prepared.

0.6 M Hydrochloric acid (HCl) (Shelf Life: 6 months at 4 °C)

For 1000 ml of 0.6 M HCl, add:

- 50 ml of 37% HCl (~ 12 M stock solution) to 950 ml ultrapure water or
- 100 ml of 20% HCl (~6 M stock solution) to 900 ml ultrapure water

Mix by gently inverting. **!ATTENTION! Always put the water first, then add the acid slowly!** Store the finished solution at 4°C and mix by hand shaking before use.

0.1 M Sodium Hydroxide (NaOH) (Shelf Life: 6 months)

For 1000 ml of 0.1 M NaOH, dissolve 4g of stock NaOH pellets (~ 2 pellets) in 1000 ml ultrapure water. Mix by gently inverting. Store at 4 °C and mix by shaking before use.

50 mM Ammonium Bicarbonate buffer, pH 8 (Shelf Life: 6 months)

For 1000 ml of 50 mM AmBic (NH_4HCO_3 , also called ammonium hydrogen carbonate), weigh out 3.95 g of AmBiC powder and transfer it into a clean glass bottle. Fill up to 1000 ml and mix by gently inverting. Check the pH and adjust if necessary. Store at 4 °C.

pH adjustment: Add a few droplets of 1 M AmBic to the prepared 50 mM AmBic and mix by shaking. Check the pH and repeat if necessary.

5% Trifluoroacetic acid (TFA) (Shelf life: 6 months)

NOTE: Trifluoroacetic acid (TFA) is very hazardous (corrosive)! Handling is only allowed under the fume hood after proper briefing! Prepare this solution in a brown glass bottle since TFA is slightly light sensitive. Always pipette water first, then the acid! Repeated use of plastic pipette tips in TFA stock solutions cause plastics to build up in the TFA and can interfere with mass spectrometry.

For 20 ml, dilute 1 ml TFA stock solution in 19 ml and mix by gently inverting.

Wash solution (Shelf Life: 6 months)

0.1% TFA in ultrapure water

For 1000 ml, add 1000 µl of TFA stock solution to 1000 ml of ultrapure water and mix by gently inverting. Store at room temperature.



Conditioning solution (Shelf Life: 3 month)

0.1% TFA in 50% Acetonitrile (ACN)

For 1000 ml, mix 500 ml ACN with 500 ml ultrapure water. Add 1000 µl of TFA carefully (see instructions above). Store at room temperature.

ACN is a volatile compound and if there is too much headspace in the storage bottle the concentration of ACN in solution goes down. Don't store small solution amounts in large bottles over a long period of time. Either prepare new conditioning solution or store the remaining solution in smaller bottles

Trypsin solution (Stable once resuspended for 1 month at -20°C)

Freeze dried Trypsin (stored at -20 °C) is resuspended in 50 µl of Trypsin resuspension buffer (included in Trypsin kit). The final concentration is now 0.4 µg/µl. Transfer the solution in a labelled (content, date, name) 0.5 ml tube and store at -20°C.



II SECTION: SAMPLE PREPARATION

1. Weighing out the sample

1.1 Archaeological samples:

Weigh 10-20 mg of bone chips or ~10 mg bone powder into a microcentrifuge tube labelled with the sample name, date and your name. Record the weight.

1.2 Modern samples:

Weigh 5-15 mg of bone chips or ~7 mg bone powder into a microcentrifuge tube labelled with the sample name, date and your name. Record the weight.

2. Collagen Extraction – Protocol selection

You can analyse the sample non-destructively (buffer extraction) or destructively (acid pretreatment).

With the destructive method you can analyse the acid soluble collagen fraction (acid supernatant), the acid insoluble collagen fraction (bone chip fraction), or both from the same sample.

For:

Nondestructive analysis (buffer extraction):

Section A (III, Chapter 3) then continue from Section Digestion and Peptide Extraction (VI, Chapter 6)

Destructive analysis (acid pretreatment):

Acid insoluble collagen fraction:

Section B (IV, Chapter 4) and then continue from Section Digestion and Peptide Extraction (IV, Chapter 6)

Acid soluble collagen fraction:

Section C (V, Chapter 5) then continue from Section Digestion and Peptide Extraction (IV, Chapter 6)

Both Acid soluble and insoluble fractions:

Section B (IV, Chapter 4) and C (V, Chapter 5) then from Section Digestion and Peptide Extraction (IV, Chapter 6)



III SECTION A: NONDESTRUCTIVE COLLAGEN EXTRACTION

3. Buffer Extraction

3.1 Pretreatment

- Add 100µL of 50 mM AmBic to the sample. If large, use enough AmBic to cover the whole sample. Leave sample at ambient temperature overnight (or for up to a week) to clean and remove any soluble contamination.
- Centrifuge sample at high speed for 1 min. Remove and discard the supernatant.

NOTE: If the sample is high precious or heavily degraded, you can lower the centrifuge speed or skip this step to avoid sample destruction.

3.2 Gelatinization

- Add 100 µL of AmBic to sample. Incubate for 1 hour at 65°C.

NOTE: If your sample amount is limited or if less sample than recommended was available, use 50 µl of AmBic for gelatinization. If your bone sample is large, use as much AmBic as needed to cover the whole bone for gelatinization.

- During incubation, label two new sets of microcentrifuge tubes. One set should be labelled with “EXT” (extract) and sample name and the second set with “SE” (second extract) and the sample name. If you used 50µl of AmBic for gelatinization, only the EXT tube is needed.
- Centrifuge your samples for 1 min after incubation.
- Transfer now 50 µl of the supernatant to the “EXT” labelled tube and the remaining 50 µl supernatant to the “SE” labelled tube.

NOTE: If the original bone sample needs to be stored, dry the bone/sample at room temperature (please see protocol for drying samples/long-term storage).

NOTE: If the bone was large and required more than about 200 µl of AmBic to cover it, the volume will need to be reduced by evaporation (either open tubes on a heat block or a vacuum centrifuge) until it is a suitable volume (usually around 100-200 µl to continue with the protocol as written).

Go to Procedure Section: Digestion and Peptide Extraction (IV, Chapter 6).



IV SECTION B: DESTRUCTIVE ANALYSIS/ACID INSOLUBLE COLLAGEN

4. Bone Chip/Bone Powder Fraction

4.1 Acid Demineralization (destructive)

- Add 500 μL of cold (4°C) 0.6 M HCl to the sample (or enough acid to cover the sample completely) and place the tubes in the fridge at 4°C . Leave until demineralization is complete (bone becomes flexible / spongy and no bubbling is visible; this may take 1 day to 2 weeks for specific sample types). Less time will be needed for bone powders (4-10h).

NOTE: For samples with good collagen preservation, complete demineralization is not necessary. Please be aware that the more the collagen remains in acid the more it breaks down. Hence monitor samples carefully at this stage to reduce protein loss.

- Centrifuge the sample for 1 min to settle the bone and (heavier) collagen to the base of the tube. Remove the acid supernatant to a new tube labelled with the sample name and "AC" and store at -20°C . *!Do not disturb your sample during pipetting!*

NOTE: If you want to analyse the acid soluble collagen fraction you should use this AC supernatant and go to SECTION C (V, Chapter 5: Acid soluble collagen).

4.2 Washing and Humic Acid Removal

- Add 200 μL of 50 mM AmBic, briefly mix by pipetting or inverting the sample, centrifuge, and then discard. Repeat for a total of 3 times.

NOTE: This step is to wash the sample back to neutral pH. If the sample is very small or there is little collagen, less washes can be performed to avoid collagen loss.

- Add 200 μL of 0.1 M NaOH, mix by pipetting or inverting the sample, and incubate for 5 min. Centrifuge for 1 min. Remove and discard NaOH wash.

NOTE: This step is performed to remove humic acids which interfere with the MALDI TOF MS. If the sample is believed to be highly degraded already, this step can be skipped as it may further damage the remaining collagen.

- Add 200 μL of 50 mM AmBic, briefly mix by pipetting or inverting the sample, centrifuge, and then discard. Repeat for a total of 3 times



NOTE: This step is to wash the sample back to neutral pH after NaOH washing. If you skipped that step, you can skip those washes. If the sample is very small or there is little collagen, less washes can be performed to avoid collagen loss.

4.3 Gelatinization

- Add 100 µl of AmBic to sample. Incubate for 1 hour at 65°C.

NOTE: If your sample amount is limited or if less sample then recommended was available, use 50 µl of AmBic for gelatinization. If your bone sample is large, use as much AmBic as needed to cover the whole bone for gelatinization.

- During incubation, label two new sets of microcentrifuge tubes. One set shall be labelled with “EXT” (extract) and sample name and the second set with “SE” (second extract) and the sample name. If you used 50 µl of AmBic for gelatinization, only the EXT tube is needed.
- Centrifuge your samples at 20000 rpm (or at highest speed) at room temperature for 1 min after incubation.

NOTE: If the sample is high precious or heavily degraded, you can use a lower centrifuge speed to avoid sample destruction.

- Transfer now 50 µl of the supernatant to the “EXT” labelled tube and the remaining 50 µl supernatant to the “SE” labelled tube.

NOTE: The “SE” labelled tube serves as back up and can be stored at -20 °C. Adjust the amounts of “EXT” and “SE” if less AmBic was used for gelatinization.

NOTE: If the bone was large and required more than about 200 µl of AmBic to cover it, the volume will need to be reduced by evaporation (either open tubes on a heat block or a vacuum centrifug) until it is a suitable volume (usually around 100 µl - 200 µl) to continue with the protocol as written.

NOTE: If the original bone sample needs to be stored, freeze sample at -20°C or dry the sample at room temperature (please see protocol for drying samples/long-term storage).

Go to Procedure Section: Digestion and Peptide Extraction (IV, Chapter 6).



V SECTION C: DESTRUCTIVE ANALYSIS/ACID SOLUBLE COLLAGEN

5. Acid Supernatant

5.1 Acid Demineralization (destructive)

- Add 500 μ L of cold (4 °C) 0.6 M HCl to the bone chip sample (or enough acid to cover the sample completely) and place the tubes in the fridge at 4 °C. Leave until demineralization is complete (bone becomes flexible/spongy and no bubbling is visible; this may take 1 day to 2 weeks for specific sample types). Less time will be needed for bone powders (4-10 h).

NOTE: For samples with good collagen preservation, complete demineralization is not necessary. Please be aware that the more the collagen remains in acid the more it breaks down. Hence monitor samples carefully at this stage to reduce protein loss.

- Centrifuge the sample for 1 min to settle the bone and (heavier) collagen to the base of the tube. Remove the acid supernatant into a new tube labelled with the sample name and "AC". !ATTENTION! Do not disturb your sample during pipetting!

NOTE: If you want to analyze the acid insoluble collagen fraction you should use this bone chip/bone powder fraction and go to SECTION B (IV, Chapter 4: Acid insoluble collagen).

5.2 Buffer exchange

- Place 30 kDa ultrafilters into labelled tubes.
- Take 250 μ L of acid supernatant and put into a 30 kDa ultrafilter. Take care not to touch the tip to the membrane.

NOTE: For well-preserved collagen samples a 30 kDa ultrafilter is sufficient. Smaller sized filters easily get clogged. For poorly preserved collagen, a 3 kDa or 10 kDa ultrafilter can be considered.

- Put on cap and place filters into the centrifuge with the cap strap toward the center of rotor, this ensures the special shaped ultrafilters will fit properly into the centrifuge. Centrifuge at 3700 rpm until acid has been pushed through the filter. This can take between 5 and 30 min.

NOTE: You should not let the filter dry out during this stage. Small amounts of liquid above the filter are fine.



- Discard flow-through
- Add 500 μ L of 50 mM AmBic and centrifuge at 3700 rpm until AmBic has passed through filter.
- Discard flow-through
- Add 100 μ L of 50 mM AmBic to the filter and mix carefully by pipetting.

NOTE: If your sample amount is limited or if less sample than recommended was available, use 50 μ L of AmBic.

- Label two new sets of tubes. One set shall be labelled with “**EXT**” (extract) and sample name and the second set with “**SE**” (second extract) and the sample name. If you are going to use 50 μ L of AmBic for eluting the collagen from the filter, only the EXT tube is needed.
- Transfer 50 μ L of the supernatant to the “EXT” labelled tube and the remaining 50 μ L supernatant to the “SE” labelled tube. Discard the filter.

Go to Procedure Section: Digestion and Peptide Extraction (IV, Chapter 6).



VI SECTION: DIGESTION AND PEPTIDE EXTRACTION

6. Trypsin Digestion

- Add 1 μL of trypsin solution (0.4 $\mu\text{L}/\mu\text{g}$) to the “EXT” labelled tube. Incubate sample overnight at 37 °C (approx. 12-18 h).
- Centrifuge sample for 1 min and add 1 μL of 5% TFA to stop the trypsin. The peptides can now be purified using a C18 ZipTip (see Chapter 6.0). If not extracting immediately, store the samples in the freezer at -20°C at this point.

7. Peptide Purification

7.1 Preparation

For each sample you must prepare separate tubes of wash and conditioning solution. Prepare as follows:

- “C” labelled tube - 250 μL of Conditioning Solution (or you can just use a common reagent reservoir)
- “W” labelled tube - 450 μL of Washing Solution
- New tube labelled with sample name and “COL” – 50 μL Conditioning Solution.

Note: During your extraction your peptides are bound to the C18 filter sitting in the pipette tip. The peptides then need to be washed and eluted. Therefore, subsequent wash and conditioning solutions will be ‘contaminated’ with your sample and so solutions after the sample has been bound to the C18 filter and cannot be used for more than one sample.

7.2 C18 ZipTipping

*Note: Before use, each tip needs to be conditioned. A separate tip is used for each standard and sample. The pipette should be set at 100 μL and the volume NOT changed during the time. Be careful to avoid pulling air through the filter, especially at the steps with only 50 μL of solution. Use the **ZipTipping Only** pipettes as repeated use of the filters pulls the pipettes out of calibration.*



Purify your peptides using the C18 ZipTip from the digested “EXT” sample as follows:

- Rinse tip twice with 100 μ L of conditioning solution and discard.
- Rinse tip twice with 100 μ L of wash solution and discard.
- Resuspend the sample back and forth over the tip, at least 5 times. !Be careful not to push air through the C18 filter (never go past the first stop point of the pipette)!
- Rinse tip twice with 100 μ L of wash solution and discard.
- Elute sample into the prepared “COL” labelled tube with the prepared 50 μ L of conditioning solution and pass through tip at least 5 times.

The sample(s) are now ready to be spotted on the MALDI plate. If not spotting immediately, store sample(s) in freezer at -20°C.

COL tubes should be taken down to the MALDI lab and should be spotted there and then stored in the freezer in the MALDI lab. All other tubes should be stored in the ZooMS Wet Chemistry Lab.